AALTOES × DESIGN FACTORY

Autonomous courier robot for transporting coffee mugs

- We're building a robot from scratch — from chassis to software.
- It will autonomously pick up mugs from Startup Sauna and deliver them to Kafis.
- The project is a collaboration between Aaltoes and Design Factory. If this experiment succeeds, the robot will stay at Design Factory as a living example of what gets made here.

Example of a delivery robot from Oodi library

> The goal is not just to build a prototype — but to run a working experiment in a real environment and show what student projects are capable of.

WHAT THE ROBOT DOES

Basic scenario: pick up → deliver → return

Step 1

Drives to the station,
picks up the crate with mugs

Step 2

Autonomously drives to Kafis
along the planned route

Step 3

Leaves the crate and
returns to home point

Perception & Navigation

LiDAR + depth camera, SLAM — the robot builds a map and finds its path autonomously

Control

Telegram bot or app — launch the route and track the robot's status in real time

TECH STACK — 5 DIRECTIONS

Everyone finds their role

Mechanics

Chassis, crate pickup mechanism

Electronics

Power, wiring, components

Embedded

Jetson Orin Nano, ESP32 / STM32, low-level control

Perception / Nav

LiDAR, camera, SLAM on ROS 2

Backend / UX

Server, Telegram bot, communication with the robot

> Working in teams by stack. Each group has its own area of responsibility, a shared chat, and regular meetings with the full project.

Schedule

Kickoff — May. Main work — summer.

Equipment

Organizers provide everything needed. Participants just need to build.

Venue

Design Factory + Electrical Lab of the EE department. Access to tools and labs.

Result

A working prototype in a real environment.

GET INVOLVED

Want to build the robot?

Application form — join the project

The project is led by Viktor · Telegram: t.me/srviktorian